Externalities

Identify the spillover → explain the private decision → find the efficient outcome → evaluate solutions

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1. What is an externality?

In a competitive market, buyers compare the benefits of a purchase with its price, and firms compare the revenue from selling with their production costs. When these decisions account for *all* benefits and costs, the market can allocate resources efficiently. But a transaction sometimes affects people who are not buying or selling the good. Then private decisions can leave out something important.

> **DEFINITION Externality**
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> An **externality** is a cost or benefit of an activity that falls on someone else and is not fully taken into account by the person making the decision. The person affected is sometimes called a **bystander**. An externality is a potential source of **market failure**: private choices need not maximize total benefits minus total opportunity costs.

Consider a business arranging overnight deliveries along a residential street. The buyers enjoy getting their orders promptly, and the delivery company pays its drivers and fuel costs. Nearby residents also hear the trucks. If nobody compensates them for this disturbance, the delivery decision may ignore a real cost to others. That is a **negative externality**.

Now consider a worker taking a first-aid course. She benefits from knowing how to respond in an emergency. Her coworkers and other people she might help also benefit, even though they did not pay for the course. If she considers only her own benefit, that is a **positive externality**.

> **KEY RESULT The direction of the market failure**
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> **Negative externality:** a decision imposes an unaccounted-for cost on others. The market typically produces **too much of the activity** relative to the efficient amount.
> **Positive externality:** a decision creates an unaccounted-for benefit for others. The market typically produces **too little of the activity** relative to the efficient amount.
> These statements assume the usual competitive-market model and no other distortions. They concern *quantities*, not whether the entire activity is good or bad.

Why does this matter? A buyer and seller can both gain from a transaction while imposing costs on somebody else. Equally, someone can produce benefits for others without being paid for them. To judge efficiency, economists consider *everybody affected*, not just the people making the transaction.

2. Negative externalities: why can the market produce too much?

DEFINITION Private cost, external cost, and social cost

Private marginal cost (PMC) is the extra cost paid by the producer for one more unit. **Marginal external cost (MEC)** is the extra harm that unit imposes on others. **Social marginal cost (SMC)** is their sum:

$$\boxed{\text{Social marginal cost} = \text{private marginal cost} + \text{external cost.}}$$

Marginal benefit (MB) is what buyers gain from one more unit. *Marginal* always means *one additional unit*.

Worked example: Overnight delivery runs

Suppose each additional overnight delivery run gives customers a benefit, costs the delivery company money, and causes $8 of disturbance to nearby households. The numbers below are *hypothetical marginal values* at different quantities of deliveries. They describe the benefit and costs of additional activity, not the totals for all deliveries.

Quantity Q	Marginal benefit	Company cost	Cost to residents	Full social cost
6	$24	$12	$8	**$20**
8	$22	$14	$8	**$22**
10	$20	$16	$8	**$24**
12	$18	$18	$8	**$26**

The private decision. The delivery company and its customers compare the benefit of another run with the company's cost. At 12 runs, both are $18, so the private market stops there. Neither side pays the nearby households for the disturbance.

The social decision. At 10 runs, the next run creates $20 of benefit but uses resources costing $16 and inflicts $8 of disturbance: $24 in total. That run *reduces* social surplus by $4. At 8 runs, the benefit and full social cost are both $22. This is where an efficient choice stops in our smooth illustration.

KEY RESULT The efficient-quantity rule

Produce additional units while their **benefit to society** is greater than their **full social cost**. Stop where the two are equal (or, for indivisible units, before the first additional unit whose social cost exceeds its benefit). With a negative production externality, the efficient quantity is typically **smaller** than the unregulated market quantity.

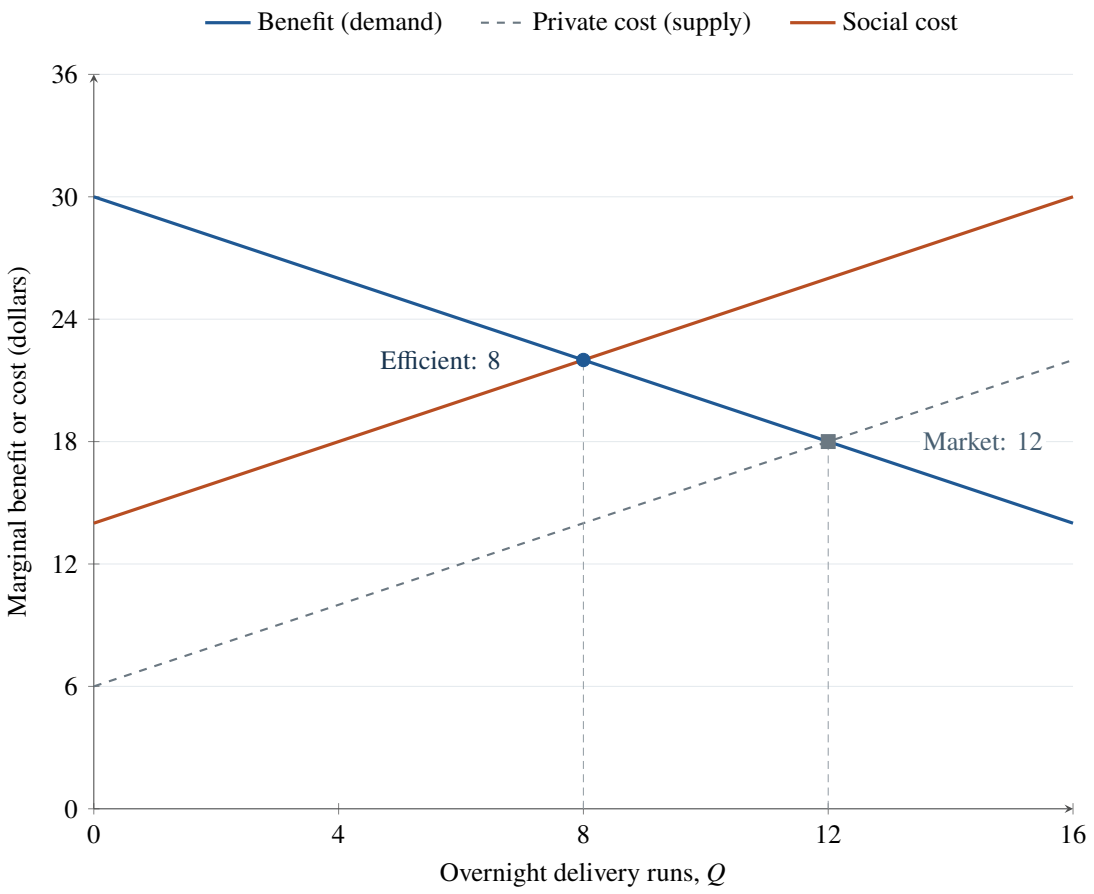


Figure 1. Negative externality. The market uses the lower, private-cost curve and chooses 12 runs. Including the $8 harm to residents raises the cost to society; 8 runs are efficient.

Read the graph. The gray dashed line is the cost the company pays; the orange line adds the cost imposed on residents. The **vertical gap is $8 at every quantity**. The demand curve crosses private cost at 12 but social cost at 8. Between 8 and 12, the extra deliveries cost society more than the customers value them. **That is the inefficiency, not merely the existence of noise.**

3. A corrective tax: making decision makers face the harm

> **DEFINITION Internalizing an externality**
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> To **internalize an externality** is to change incentives so that a person making a decision takes account of the external cost or benefit. A **corrective (Pigouvian) tax** places a charge on an activity that imposes a negative external cost.

Return to the delivery example. Suppose the city charges $8 per overnight run. A run that costs the company $14 in resources now also requires an $8 tax payment. For the company's decision, the relevant cost is $22—the full social cost at 8 runs. Buyers and sellers now choose 8 runs instead of 12.

KEY RESULT Why a corrective tax can improve efficiency

In the simple model, a tax equal to the **marginal external cost at the efficient quantity** aligns private incentives with social cost. Here an \$8 tax per run changes the private choice from 12 runs to 8. **The tax discourages the deliveries whose benefits do not justify their full social costs.**

Where does the tax go? The tax is paid to the government; it is not itself the noise damage. The real social loss is the disturbance. The payment changes the incentive to make deliveries. It does not compensate residents automatically or eliminate all deliveries. Whether buyers or sellers are legally required to remit the tax does not change the basic quantity result in this competitive model, although who bears the burden depends on supply and demand.

Why not prohibit every overnight delivery? At 6 runs, an additional run gives customers \$24 of benefit and has a full social cost of only \$20. Eliminating that worthwhile activity would destroy gains from trade. **The efficient goal is to account for the harm, not necessarily to eliminate the activity.**

4. Positive externalities: why can the market produce too little?

A positive externality reverses the comparison. Someone undertaking an activity enjoys a private benefit, but other people gain as well. The decision maker may not receive payment for those extra benefits.

DEFINITION Private, external, and social marginal benefit

Private marginal benefit (PMB) is the additional benefit to the person making a choice. **Marginal external benefit (MEB)** is the additional benefit received by others. **Social marginal benefit (SMB)** includes both:

$$\text{Social marginal benefit} = \text{private benefit} + \text{external benefit}.$$

Worked example: First-aid training places

A provider offers short first-aid courses to employees. The trainee gains a useful skill; coworkers and members of the public may also benefit if that person helps during an emergency. Suppose each additional course place creates \$6 of benefit for people other than the trainee. The costs and benefits shown are hypothetical.

Quantity Q	Trainee's benefit	Benefit to others	Total benefit	Marginal cost
6	\$16	\$6	**\$22**	\$10
9	\$13	\$6	**\$19**	\$13
12	\$10	\$6	**\$16**	\$16
14	\$8	\$6	**\$14**	\$18

The private decision. At 9 places, the next trainee values training at \$13 and providing a place costs \$13. Beyond that, the trainee's private willingness to pay is lower than the cost, so the market stops at 9.

The social decision. At 9 places, the total benefit is \$13 to the trainee plus \$6 to other people, or \$19; the place costs only \$13. More training would increase total surplus. At 12 places, total benefit and cost are both \$16. Thus 12 places, not 9, are efficient in this illustration.

KEY RESULT Positive externality: the efficient quantity exceeds the market quantity

The market compares private benefit with cost. Society compares **private plus external benefit** with cost. When the external benefit is ignored, some worthwhile units are not purchased. The result can be **underprovision** even though buyers and sellers are acting in their own interests.

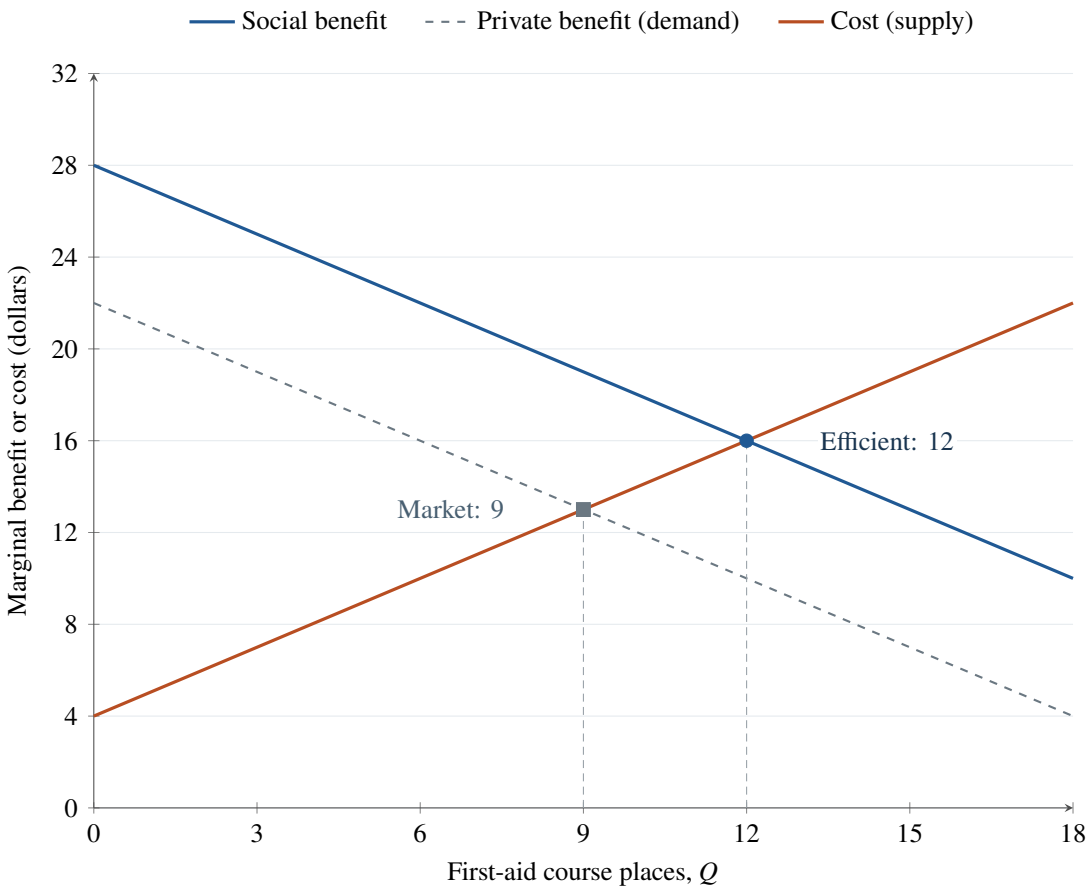


Figure 2. Positive externality. The social-benefit curve lies $6 above trainees' private benefit. The private market supplies 9 places; allowing for benefits to others makes 12 places efficient.

Read the graph. The dashed gray curve shows what trainees themselves are willing to pay. The blue line includes the $6 benefit that others receive, so it is **above** private demand. Between 9 and 12, the total benefit of additional training exceeds its cost, but trainees will not pay the full amount. **These missing gains explain the underprovision.**

DEFINITION Corrective subsidy

A **corrective subsidy** is a payment that encourages an activity with external benefits. An ideal per-unit subsidy in the simple model equals the marginal external benefit at the efficient quantity.

A $6 subsidy closes the gap. At 12 places, the provider receives $16: the trainee pays $10 and taxpayers contribute $6. Private benefit ($10) plus benefit to others ($6) equals the $16 resource cost. **The subsidy gives trainees an incentive to purchase socially worthwhile places.** In practice, financing the subsidy also has a cost.

5. Public policies: regulation, taxes, and tradable permits

Policy choice depends on the harm, the information available, and enforcement costs.

DEFINITION Three main policy instruments

Command-and-control regulation sets direct limits or methods, such as an emissions ceiling or required filter.

A **corrective tax** charges each unit of harmful activity. A corrective **subsidy** rewards activity with external benefits.

A **tradable-permit system** caps total emissions. Firms need permits to emit, but can trade them.

KEY RESULT Price versus quantity

Tax: sets a price per unit of emissions; firms determine how much to emit.

Tradable permits: set total emissions; the market determines the permit price.

Direct regulation: sets a required limit or method. The instruments differ in costs, incentives, and information needed.

Worked example: Why trading pollution permits can save resources

Two workshops each emit 10 units of smoke per day. The authority sets a total cap of **12**, requiring 8 units of reduction. Workshop A can cut smoke for $10 per unit; B's cost is $30. Assume these costs stay constant.

	A cuts	B cuts	Total real cost
Equal cuts: 4 units each	4 at $10 = $40	4 at $30 = $120	**$160**
Least-cost cuts: A does all 8	8 at $10 = $80	0 = $0	**$80**

Why are equal cuts costly? One unit of reduction costs B $30 but A only $10. Shifting a unit of cleanup from B to A saves $20 with no change in emissions. Repeating that trade lowers the total cost.

Give each workshop 6 tradable permits, allowing one unit of smoke per permit. A cuts 8, emits 2, and sells its **4 unused permits** to B. B holds 10 permits and makes no cuts. Emissions remain **2 + 10 = 12**, but cleanup costs just $80. At a permit price between $10 and $30, both prefer to trade.

KEY RESULT Why tradable permits can be less costly

A permit market encourages firms that can clean up **cheaply** to do more of the cleanup, while firms that face **high costs** buy permits. For a given properly enforced cap, this can achieve the required reduction at lower total cost than requiring equal cuts from every firm. **The permits do not make pollution harmless; they change who reduces it.**

Three qualifications. An emissions cap must be chosen: trading alone does not tell us the socially efficient level of pollution. Permit markets need monitoring and enforcement. And the saving from trade in this example relies on the assumed costs and the ability to trade; different facts can change the result.

6. Private solutions: bargaining and the Coase theorem

Not every externality requires a new government tax or regulation. When the people affected can identify each other and reach enforceable agreements, they may have incentives to correct the problem themselves.

> **DEFINITION Coase theorem**
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> If property rights are clear, parties can bargain freely, and bargaining and enforcement have **no transaction costs**, they can reach an efficient allocation despite an externality. Under those conditions, the initial assignment of rights affects *who pays whom*, but not which activity is efficient.

Worked example: Two businesses and a shared loading area

A wholesale bakery saves $240 each evening by using a loading bay at a particular hour. Its trucks block the adjacent florist's deliveries, causing the florist $400 in lost business. Suppose the bakery has the legal right to use the bay.

What would happen without bargaining? The bakery uses the bay because it gains $240 and does not pay the florist's $400 loss. But this activity causes a $160 net loss to the two firms together ($400 harm minus $240 gain). It is inefficient.

Can the firms reach an agreement? Yes, if bargaining is costless. The florist could pay the bakery $300 to change its delivery hour. The bakery loses the $240 saving but receives $300, so it gains $60. The florist avoids $400 of loss and pays $300, so it gains $100. Together they gain $160, the value of avoiding the inefficient arrangement.

What if the florist has the right to keep the bay clear? The bakery would change its hour without payment. The efficient result is still a clear bay; the allocation of the payment changes. In both cases, rights matter for distribution, while the efficient activity is determined by comparing the $240 benefit with the $400 harm.

> **KEY RESULT Why private bargaining may fail**
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> **Transaction costs** include finding the parties, negotiating, writing and enforcing agreements. Bargaining may fail if these costs exceed the potential gain ($160 in this example), if rights are uncertain, or if many affected people cannot coordinate. **The Coase theorem is a benchmark showing when voluntary bargaining can work, not a promise that bargaining always works.**

7. How to reason through an externality question

> **KEY RESULT A reliable method**
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> **1. Identify the activity.** What extra unit is produced or consumed? Who makes the decision?
> **2. Identify the bystander.** Who bears a cost or receives a benefit without it being reflected in the decision?
> **3. Compare private and social incentives.** Negative external cost: social cost exceeds private cost, so the market tends to produce too much. Positive external benefit: social benefit exceeds private benefit, so the market tends to produce too little.
> **4. Apply the efficiency rule.** An additional unit is worthwhile only if its **total social benefit** is at least its **total social cost**.
> **5. Explain the remedy.** Ask precisely what the tax, subsidy, rule, permit, or agreement changes about the private decision.

Do not confuse three claims. *An externality exists* does not automatically mean the efficient amount of the activity is zero. *A policy reduces pollution* does not establish that it does so at least cost. And *a government intervenes* does not by itself establish that the intervention improves total surplus. The economic question is always what changes, why, and at what opportunity cost.

Worked example: Two more applications of the rule

A household hires a powered leaf blower that fills the neighbors' outdoor seating area with dust. The homeowner values a tidy yard, but neither the homeowner nor the service supplier necessarily pays for the disruption. **That is a negative externality:** if the harm is ignored, too much noisy cleaning may be purchased. A rule limiting operating hours, a charge reflecting the harm, or an agreement with neighbors can reduce the mismatch.

Suppose a small employer pays to teach staff a software skill, and some of the trained workers later share those skills with other businesses. If the original employer cannot capture those additional benefits, it may invest in too little training. **That is a positive externality:** some of the return goes to people outside the original decision. A well-targeted subsidy or a voluntary cost-sharing arrangement can help.

These examples differ from the delivery and first-aid cases, but the reasoning is unchanged: **identify whose benefit or cost is missing, then show how the missing incentive affects the quantity chosen.**